

Midterm Exam

Time: Mar/29/25 Sat 9:30am - 11:30am

DURATION OF EXAMINATION: 2 hours

Your exam sheet shall include 6 problems. If not, notify the instructors.

1. (18 pt) Solving a Linear System of Equations

Consider the system of linear equations:

$$\begin{cases} x_1 + 2x_2 - x_3 + 4x_4 = 3 \\ 2x_1 - x_2 + 3x_3 + 5x_4 = -1 \\ 3x_1 + 4x_2 + 2x_3 - x_4 = 2 \end{cases}$$

(a) (3 pt) Write the coefficient matrix A and the corresponding augmented matrix.

Solution: The coefficient matrix A and the augmented matrix $[A|b]$ are given by:

$$A = \begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & -1 & 3 & 5 \\ 3 & 4 & 2 & -1 \end{pmatrix}, \quad [A|b] = \begin{pmatrix} 1 & 2 & -1 & 4 & 3 \\ 2 & -1 & 3 & 5 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}$$

(b) (5 pt) Compute the row-reduced echelon form (RREF) of the augmented matrix.

Solution: Performing row operations to obtain the row-reduced echelon form:

$$\begin{aligned} & \begin{pmatrix} 1 & 2 & -1 & 4 & 3 \\ 2 & -1 & 3 & 5 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & -1 & 4 & 3 \\ 0 & -5 & 5 & -3 & -7 \\ 0 & -2 & 5 & -13 & -7 \end{pmatrix} \\ & \rightarrow \begin{pmatrix} 1 & 0 & 1 & \frac{14}{5} & \frac{1}{5} \\ 0 & 1 & -1 & \frac{13}{5} & -\frac{7}{5} \\ 0 & 0 & 3 & -\frac{59}{5} & -\frac{21}{5} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & \frac{101}{15} & \frac{8}{5} \\ 0 & 1 & 0 & -\frac{10}{15} & 0 \\ 0 & 0 & 1 & -\frac{59}{15} & -\frac{7}{5} \end{pmatrix} \end{aligned}$$

(c) (5 pt) Express the solution set of the system.

Solution: Expressing the solution:

$$\begin{aligned} x_1 &= -\frac{101}{15}x_4 + \frac{8}{5} \\ x_2 &= \frac{10}{3}x_4 \\ x_3 &= \frac{59}{15}x_4 - \frac{7}{5} \end{aligned}$$

Thus, the parametric solution is:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -\frac{101}{15} \\ \frac{10}{3} \\ \frac{59}{15} \\ 1 \end{pmatrix} x_4 + \begin{pmatrix} \frac{8}{5} \\ 0 \\ -\frac{7}{5} \\ 0 \end{pmatrix}$$

(d) (3 pt) Determine the rank and nullity (i.e., dimension of null space) of the coefficient matrix A .

Solution: From the definition, $\text{rank}(A) = 3$. Since A has 4 columns, the nullity is:

$$\text{nullity}(A) = 4 - 3 = 1$$

(e) (2 pt) Find a basis for the null space of A . (no need to prove it is a basis)

Solution: A basis for the null space consists of the coefficient of the free variable vector:

$$\left\{ \begin{pmatrix} -\frac{101}{15} \\ \frac{10}{3} \\ \frac{59}{15} \\ 1 \end{pmatrix} \right\}$$

2. (15 pt) Question Answering

(a) (3 pt) Is $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 + x_2 = 1\}$ a linear space? Justify your answer.

(b) (3 pt) Suppose $u, v \in \mathbb{R}^{n \times 1}$, and $A, B, C \in \mathbb{R}^{n \times n}$. Which of the following matrix products are valid? $uAv, u^T ABv, ABCvuA, u^T Avu^T BCvv^T A$. Briefly justify your answer.

(c) (3 pt) Suppose a matrix $A \in \mathbb{R}^{5 \times 8}$ and $\text{rank}(A) = 3$. Find $\dim(N(A))$ and $\dim(N(A^T))$. Briefly explain your calculations.

Reminder: For any matrix Z , $N(Z)$ denotes the null space of Z .

(d) (3 pt) State a necessary and sufficient condition for a diagonal matrix $A \in \mathbb{R}^{n \times n}$ to be invertible, and provide a concise proof.

(e) (3 pt) Suppose $A \in \mathbb{R}^{n \times n}$. Is it possible to have a matrix A where $A \neq I$ and $A \neq 0$ such that $A^2 = A$? State your answer and briefly explain why.

Solution:

(a): Answer(1 pt): No.

Reason(2 pt): The answer is deemed correct if the student accurately identifies and explains at least one key aspect.

1. Closure under addition(1 pt):

Let $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ be vectors in the set $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 + x_2 = 1\}$. Then $x_1 + x_2 = 1$ and $y_1 + y_2 = 1$. The sum of these two vectors is $\mathbf{x} + \mathbf{y} = (x_1 + y_1, x_2 + y_2)$. Calculating $(x_1 + y_1) + (x_2 + y_2) = (x_1 + x_2) + (y_1 + y_2) = 1 + 1 = 2 \neq 1$. So, S is not closed under addition.

2. Closure under scalar multiplication(1 pt):

Let $\mathbf{x} = (x_1, x_2) \in S$, so $x_1 + x_2 = 1$, and let $k \in \mathbb{R}$. Then $k\mathbf{x} = (kx_1, kx_2)$. Since $kx_1 + kx_2 = k(x_1 + x_2) = k$, when $k \neq 1$, $kx_1 + kx_2 \neq 1$. Thus, S is not closed under scalar multiplication.

3. Existence of zero vector(1 pt): For a set to be a linear space, the zero vector $\mathbf{0} = (0, 0)$ must be in it. However, $0 + 0 = 0 \neq 1$, so $\mathbf{0} \notin S$.

In conclusion, the set $\{(x_1, x_2) \in \mathbb{R}^2 | x_1 + x_2 = 1\}$ does not satisfy the requirements to be a linear space, and hence it is not a linear space. (1 pt)

(b):

Answer(2 pt): $u^\top ABv$ and $u^\top Avu^\top BCvv^\top A$.

Reason(1 pt): Recall the rules of matrix multiplication: If M is an $m \times p$ matrix and N is a $q \times r$ matrix, the product MN is well - defined if and only if $p = q$, and the resulting matrix MN has size $m \times r$.

(c):

Answer(1 pt): $\dim(N(A)) = 5$, $\dim(N(A^\top)) = 2$

Reason(2 pt): By the Fundamental Theorem of Linear Algebra, for $A \in \mathbb{R}^{m \times n}$, $\dim(N(A)) = n - \text{rank}(A)$ and $\dim(N(A^\top)) = m - \text{rank}(A)$. Here, A is 5×8 with $\text{rank}(A) = 3$, so $\dim(N(A)) = 8 - 3 = 5$ and $\dim(N(A^\top)) = 5 - 3 = 2$.

(d):

Answer(1 pt): All diagonal elements are non- zero.

Reason(2 pt): An $n \times n$ matrix A is invertible if and only if $\text{rank}(A) = n$. For a triangular matrix, $\text{rank}(A)$ is the number of non- zero diagonal elements. If any diagonal element of the triangular matrix A is zero, then $\text{rank}(A) < n$, and A is not invertible.

(e):

Answer(1 pt): Yes, it is possible.

Reason(2 pt): For example, consider the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then $A^2 = A$, but $A \neq I$ and $A \neq 0$.

3. (16 pt) Matrix Transpose, Inverse and Multiplication

(a) (4 pt) Consider the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Find two different 3×2 matrices C (whose elements can only be 0, 1, or -1) such that $AC = I_2$, where $I_2 \in \mathbb{R}^{2 \times 2}$ is the identity matrix.

(b) (4 pt) Let

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}.$$

Compute AB and $B^\top A^\top$.

(c) (4 pt) Calculate

$$\begin{pmatrix} 2 & 5 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 2 & 3 \end{pmatrix}^{-1}.$$

(d) (4 pt) Show that if $A^3 = 2I$ and $B = A^2 - 2A + 2I$, then

$$B^{-1} = \frac{1}{10}(A^2 + 3A + 4I).$$

Solution:

(a)

$$C_1 = \begin{pmatrix} 1 & 1 \\ 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad C_2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(b)

$$AB = \begin{pmatrix} 1 & 3 \\ 3 & 8 \end{pmatrix}.$$

$$B^T A^T = (AB)^T = \begin{pmatrix} 1 & 3 \\ 3 & 8 \end{pmatrix}.$$

(c) The matrix is block diagonal with blocks

$$M_1 = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}.$$

Their inverses are

$$M_1^{-1} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}, \quad M_2^{-1} = \begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}.$$

Hence the inverse of the whole matrix is

$$\begin{pmatrix} 3 & -5 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & -2 & 3 \end{pmatrix}.$$

(d) Compute $B \cdot \frac{1}{10}(A^2 + 3A + 4I)$:

$$\begin{aligned} B(A^2 + 3A + 4I) &= (A^2 - 2A + 2I)(A^2 + 3A + 4I) \\ &= A^4 + 3A^3 + 4A^2 - 2A^3 - 6A^2 - 8A + 2A^2 + 6A + 8I \\ &= A^4 + (3 - 2)A^3 + (4 - 6 + 2)A^2 + (-8 + 6)A + 8I \\ &= A^4 + A^3 + 0A^2 - 2A + 8I. \end{aligned}$$

Since $A^3 = 2I$, we have $A^4 = A \cdot A^3 = 2A$. Thus

$$B(A^2 + 3A + 4I) = 2A + 2I - 2A + 8I = 10I.$$

Therefore $B \frac{1}{10}(A^2 + 3A + 4I) = I$. Similarly, $\frac{1}{10}(A^2 + 3A + 4I)B = I$. Therefore, $B^{-1} = \frac{1}{10}(A^2 + 3A + 4I)$

4. (16 pt) True or False

For each question, you do NOT need to justify. (Only writing T or F is enough)

(a) If a matrix A has full row rank, then for any vector $\mathbf{b}$, the linear system $A\mathbf{x} = \mathbf{b}$ has at most one solution.

Solution: False.

(b) If a square matrix is invertible, then the intersection of its column space and null space only contains one element which is the zero vector.

Solution: True. The null space is $\{\mathbf{0}\}$, and the column space spans $\mathbb{R}^n$.

(c) For any matrix A , the matrix $A^T A$ is symmetric.

Solution: True. $(A^T A)^T = A^T A$

(d) The intersection of two subspaces of a certain linear space must be a subspace.

Solution: True

(e) If a set of vectors in $\mathbb{R}^n$ is linearly dependent, then it cannot span $\mathbb{R}^n$.

Solution: False. Example: $\{(1, 0), (0, 1), (1, 1)\}$

(f) Any set of vectors in $\mathbb{R}^n$ containing the zero vector is linearly dependent.

Solution: True

(g) If the columns of a matrix are linearly independent, its rows must also be linearly independent.

Solution: False. For an $m \times n$ matrix with $m > n$, columns may be independent while rows are dependent.

(h) If a homogeneous linear system of equations has more variables than equations, then the system must have a nonzero solution.

Solution: True.

5. (15 pt) Matrix Multiplication

In the Llama2 architecture by Meta AI, a key component is defined as:

$$Z = ((XU) \circ (XV))W$$

where: $X \in \mathbb{R}^{n \times d}$, $U, V \in \mathbb{R}^{d \times k}$, $W \in \mathbb{R}^{k \times m}$. The open circle “ $\circ$ ” denotes the Hadamard (element-wise) product: $[A \circ B]_{ij} = A_{ij}B_{ij}$ for any two matrices $A, B \in \mathbb{R}^{n \times k}$.

- (a) (4 pt) If we multiply only the 3rd column of W by 2, while keeping other parts of W the same, what happens to the 3rd column of Z ? Justify your answer.
- (b) (4 pt) If we multiply only the 4th column of W by 2, while keeping other parts of W the same, what happens to the 3rd column of Z ? Justify your answer.
- (c) (4 pt) If we multiply only the 3rd row of X by 2, while keeping other parts of X the same, what happens to the 3rd row of Z ? Justify your answer.
- (d) (3 pt) If we multiply only the 4th row of X by 2, while keeping other rows the same, what happens to the 3rd row of Z ? Justify your answer.

Solution Guide For full credit, answers must:

- Show step-by-step matrix operations
- Explicitly track row-wise dependencies
- State final scaling factors (if any)

Key observations:

- The Hadamard product introduces nonlinear interactions.
- Each output row depends only on its corresponding input row.

Solution:

- (a) Scaling only the 3rd column of W by 2: Writing $Z = ((XU) \circ (XV))W$, the j -th column of Z depends solely on the j -th column of W . Hence, scaling the 3rd column of W by 2 scales the 3rd column of Z by 2.
- (b) Scaling only the 4th column of W by 2: Again, the 3rd column of Z depends only on the 3rd column of W . Changing the 4th column does not affect the 3rd column of Z ; thus the 3rd column of Z remains unchanged.
- (c) Scaling only the 3rd row of X by 2: The 3rd row of Z depends on the 3rd row of $(XU) \circ (XV)$. If we denote

$$(XU)_{3,:} \quad \text{and} \quad (XV)_{3,:},$$

then each of these row vectors is multiplied by 2. The Hadamard product of two scaled vectors is scaled by $2 \times 2 = 4$. Consequently, the entire 3rd row of Z is multiplied by 4.

- (d) Scaling only the 4th row of X by 2: Each row of Z depends only on the corresponding row of XU and XV . Changing the 4th row of X does not affect the 3rd row of Z , so the 3rd row of Z remains unchanged.

Key Observations:

- Column j of Z is determined by column j of W .
- Row i of Z is determined by row i of XU and XV .
- The Hadamard product $\circ$ amplifies changes if both factors are scaled.

6. (20 pt) Rank, Independence

- (a) (5 pt) Consider the elementary row operation of adding a multiple of one row to another $R_i \leftarrow cR_j + R_i$ ($i \neq j$). In the class, we mentioned the result: applying this operation to a matrix $A \in \mathbb{R}^{m \times n}$ is equivalent to left-multiplying A by the elementary matrix $E_{cR_j+R_i}$. Prove this result for the case $m = 5, i = 4, j = 2$.

- (b) (5 pt) Let $v_1, v_2, \dots, v_k$ be vectors in $\mathbb{R}^n$. Denote $w_1 = v_1 + \alpha v_2$, where $\alpha \in \mathbb{R}$. Prove: $\text{span}\{v_1, v_2, v_3, \dots, v_k\} = \text{span}\{w_1, v_2, v_3, \dots, v_k\}$.
- (c) (5 pt) Let $v_1, v_2, \dots, v_k$ be vectors in $\mathbb{R}^n$. Define $w_i = v_1 + v_2 + \dots + v_i$ for $i = 1, 2, \dots, k$. Prove that if the set $\{w_1, w_2, \dots, w_k\}$ is linearly independent, then the set $\{v_1, v_2, \dots, v_k\}$ is also linearly independent.
- (d) (5 pt) Prove the following statements: Suppose $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{l \times m}$, if $\text{rank}(A) = n$, and $\text{rank}(B) = m$, then $\text{rank}(BA) = n$.
Hint: There are multiple methods to prove it. One way is to consider the column form of A.

Solution:

(a) Proof:

We aim to prove that executing the elementary row operation $R_4 \leftarrow cR_2 + R_4$ on a matrix $A \in \mathbb{R}^{5 \times n}$ is equivalent to left-multiplying A by the corresponding elementary matrix $E_{cR_2+R_4}$.

Formally, let

$$A = \begin{pmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \end{pmatrix},$$

where R_k represents a $1 \times n$ row vector for $k = 1, \dots, 5$. Our objective is to demonstrate that

$$E_{cR_2+R_4}A = A' = \begin{pmatrix} R_1 \\ R_2 \\ R_3 \\ cR_2 + R_4 \\ R_5 \end{pmatrix}. (1pt)$$

The elementary matrix $E_{cR_2+R_4}$ is derived by applying the operation $R_4 \leftarrow cR_2 + R_4$ to the 5×5 identity matrix

$$I_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Upon adding c times the second row to the fourth row, we obtain

$$E_{cR_2+R_4} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & c & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. (1pt)$$

Next, we compute the product $E_{cR_2+R_4}A$. Given that $E_{cR_2+R_4}$ is 5×5 and A is $5 \times n$, we have

$$E_{cR_2+R_4}A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & c & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \end{pmatrix} = \begin{pmatrix} R_1 \\ R_2 \\ R_3 \\ cR_2 + R_4 \\ R_5 \end{pmatrix}. \quad (2 \text{ pt})$$

Since directly applying the elementary row operation to A gives $A' = \begin{pmatrix} R_1 \\ R_2 \\ R_3 \\ cR_2 + R_4 \\ R_5 \end{pmatrix}$,

which is identical to $E_{cR_2+R_4}A$, our conclusion that the two operations are equivalent holds true. (1pt)

(b) Proof:

We aim to show $\text{span}\{v_1, v_2, \dots, v_k\} = \text{span}\{w_1, v_2, \dots, v_k\}$, where $w_1 = v_1 + \alpha v_2$, $\alpha \in \mathbb{R}$.

1. Prove $\text{span}\{w_1, v_2, \dots, v_k\} \subseteq \text{span}\{v_1, v_2, \dots, v_k\}$ (2 pt):

For any $x \in \text{span}\{w_1, v_2, \dots, v_k\}$, $x = c_1 w_1 + c_2 v_2 + \dots + c_k v_k$. Substitute $w_1 = v_1 + \alpha v_2$:

$$x = c_1(v_1 + \alpha v_2) + c_2 v_2 + \dots + c_k v_k = c_1 v_1 + (c_1 \alpha + c_2) v_2 + \dots + c_k v_k.$$

Since x is a linear combination of $\{v_1, v_2, \dots, v_k\}$, $x \in \text{span}\{v_1, v_2, \dots, v_k\}$. So $\text{span}\{w_1, v_2, \dots, v_k\} \subseteq \text{span}\{v_1, v_2, \dots, v_k\}$.

2. Prove $\text{span}\{v_1, v_2, \dots, v_k\} \subseteq \text{span}\{w_1, v_2, \dots, v_k\}$ (2 pt):

For any $y \in \text{span}\{v_1, v_2, \dots, v_k\}$, $y = d_1 v_1 + d_2 v_2 + \dots + d_k v_k$. As $w_1 = v_1 + \alpha v_2$, then $v_1 = w_1 - \alpha v_2$. Substitute v_1 into y :

$$y = d_1(w_1 - \alpha v_2) + d_2 v_2 + \dots + d_k v_k = d_1 w_1 + (d_2 - d_1 \alpha) v_2 + \dots + d_k v_k.$$

Hence y is a linear combination of $\{w_1, v_2, \dots, v_k\}$, so $y \in \text{span}\{w_1, v_2, \dots, v_k\}$. Therefore $\text{span}\{v_1, v_2, \dots, v_k\} \subseteq \text{span}\{w_1, v_2, \dots, v_k\}$.

Combining both inclusions gives equality. (1pt)

(c) Proof:

Assume that $\{w_1, w_2, \dots, w_k\}$ is linearly independent. We need to show that $\{v_1, v_2, \dots, v_k\}$ is also linearly independent.

1. Express v_i in terms of w_i (1 pt): From the definition of w_i , we have:

$$v_1 = w_1, \quad v_2 = w_2 - w_1, \quad v_3 = w_3 - w_2, \quad \dots, \quad v_k = w_k - w_{k-1}.$$

2. Consider a linear combination $c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0$. Substituting the expressions for v_i , we get:

$$c_1w_1 + c_2(w_2 - w_1) + c_3(w_3 - w_2) + \cdots + c_k(w_k - w_{k-1}) = 0.$$

Rearranging terms (1 pt):

$$(c_1 - c_2)w_1 + (c_2 - c_3)w_2 + \cdots + (c_{k-1} - c_k)w_{k-1} + c_kw_k = 0.$$

3. Linear independence of $\{w_1, w_2, \dots, w_k\}$ (1 pt): Since $\{w_1, w_2, \dots, w_k\}$ is linearly independent, all coefficients must be zero:

$$c_1 - c_2 = 0, \quad c_2 - c_3 = 0, \quad \dots, \quad c_{k-1} - c_k = 0, \quad c_k = 0.$$

Starting from $c_k = 0$ and working backwards, we find:

$$c_{k-1} = 0, \quad c_{k-2} = 0, \quad \dots, \quad c_1 = 0.$$

Therefore, all coefficients $c_1, c_2, \dots, c_k$ are zero. (1pt)

4. Conclusion (1 pt): Since the only solution to $c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0$ is $c_1 = c_2 = \cdots = c_k = 0$, the set $\{v_1, v_2, \dots, v_k\}$ is linearly independent. Thus, if $\{w_1, w_2, \dots, w_k\}$ is linearly independent, then $\{v_1, v_2, \dots, v_k\}$ is also linearly independent.

(d) Proof:

Let $A = [a_1, a_2, \dots, a_n]$ be the column form, where $a_i \in \mathbb{R}^m$ for each $i = 1, 2, \dots, n$. From $\text{rank}(A) = n$, and $\text{rank}(B) = m$, both A and B have full column rank. We will show that $Ba_1, \dots, Ba_n$ are independent:

Suppose $\beta_1, \dots, \beta_n \in \mathbb{R}$ satisfy

$$\beta_1Ba_1 + \cdots + \beta_nBa_n = 0.$$

By linearity,

$$B(\beta_1a_1 + \cdots + \beta_na_n) = 0.$$

Since B has full column rank, so the only solution to $Bx = 0$ is $x = 0$ (Lecture 15 Part I). Hence

$$\beta_1a_1 + \cdots + \beta_na_n = 0.$$

Independence of $a_1, \dots, a_n$ implies $\beta_1 = \cdots = \beta_n = 0$. Thus $Ba_1, \dots, Ba_n$ are independent, which means $BA = [Ba_1, Ba_2, \dots, Ba_n]$ has full column rank.